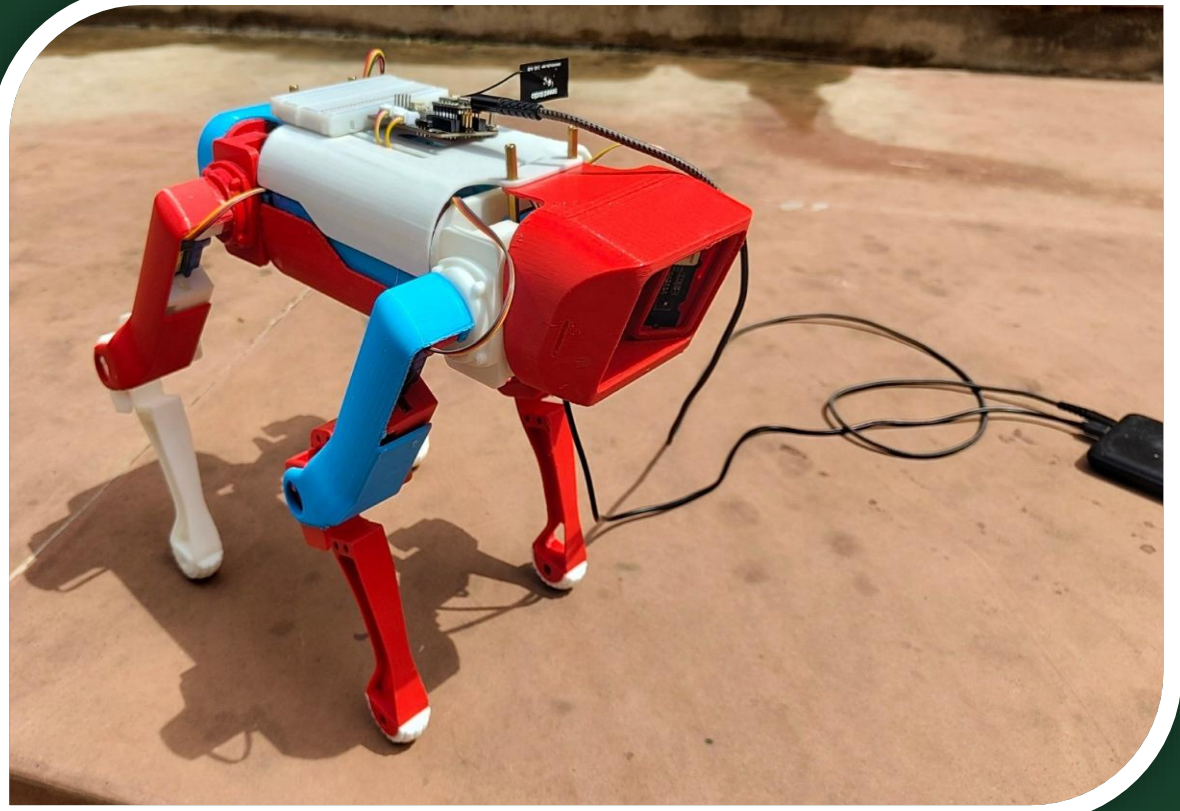
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CAPSTONE PROJECT · FINAL REVIEW

# Precision Agriculture Monitoring Quadruped Robot Dog

A 3D-printed quadruped carrying wireless crop-health sensing, with leaf-disease classification and physics-validated locomotion




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**Guide: Dr. Lakshmanan K · 54758**

# How This Review Is Organised

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| | | | |
|---|---|---|---|
| 01 | **Domain and Problem** | Crop-loss context, monitoring platforms, 18-paper survey, research gaps, objectives | Slides 3–8 |
| 02 | **Methodology and Design** | Workflow, design decisions, architecture, data flow, algorithms, locomotion model | Slides 9–19 |
| 03 | **Completion and Integration** | Scope delivered, assembled robot, integrated dashboard, stable logging, live demo | Slides 20–25 |
| 04 | **Testing and Results** | Bench tests, classifier validation, hardware actuation, simulated survey, speed limit | Slides 26–33 |
| 05 | **Innovation and Relevance** | Contributions, cost and feasibility, ethics, practical use, limitations, future work | Slides 34–40 |
| 06 | **Conclusion and Q&A** | Key takeaways, references, questions | Slides 41–43 |

# 01

# Domain and Problem



Why crop-level monitoring matters, what existing platforms miss, and what this project set out to do

# Crop Disease Is Costly, and Often Seen Too Late

**20–40 %**

of global crop production is lost to plant pests and diseases every year (FAO)

| 10–28 % | 25–41 % | 20–41 % |
|---|---|---|
| Wheat | Rice | Maize |

*Yield losses to pathogens and pests, global estimate (Savary et al., 2019)*

### Drones see only the canopy top

Short flights; no view of leaf undersides or the air between rows

### Wheeled rovers get stuck

Lose traction in mud, soft soil and narrow furrows

### Fixed IoT nodes leave gaps

Stationary sensors cover only the spot where they stand

### Research robots cost too much

Legged platforms such as ANYmal cost far beyond a small farm's budget

# Which Platform Can Monitor Between Crop Rows?

A legged robot can walk between rows and carry sensors to the leaves; the question is whether it can be built cheaply.

| Platform | Close-range, leaf-level sensing | Moves on soft soil and in furrows | Covers the whole field | Affordable for small farms |
|---|---|---|---|---|
| Drone | ✗ | ✓ | ✓ | ~ |
| Wheeled or tracked rover | ✓ | ✗ | ~ | ~ |
| Fixed IoT sensor nodes | ~ | – | ✗ | ~ |
| Research quadruped (ANYmal, Cheetah 3) | ✓ | ✓ | ✓ | ✗ |
| **This project: printed quadruped** | ✓ | ✓ | ✓ | ✓ |

✓ suited ~ partly ✗ not suited

# What 18 Papers Tell Us

## Legged locomotion [1–6]

MIT Cheetah 3, Mini Cheetah, ANYmal, Stanford Doggo; MPC and learned gaits

**Gap** Custom actuators, costly; none carries a crop-sensing payload

## Field robots and drones [7–10]

Reviews of agricultural rovers, drone–ground teaming, farmer interfaces

**Gap** Wheels fail in mud and furrows; drones cannot see under the canopy

## Leaf-disease deep learning [11–15]

CNNs reach 97–99 % on PlantVillage; MobileNetV2 is efficient enough for edge use

**Gap** Lab images only; closed-set models label anything they see

## IoT environmental sensing [16–18]

Layered IoT and sensor networks for microclimate, gases and soil

**Gap** Stationary nodes leave gaps; nothing carries sensors along rows

# Research Gaps and How This Project Answers Them

| Gap in the literature | Our response |
| --- | --- |
| Low-cost legged robots have not carried an environmental and imaging payload, and their feasibility with hobby servos is rarely quantified | A printed 12-DOF quadruped with two sensing nodes, and a physics model with the real MG996R torque limit |
| Crop sensing relies on fixed nodes; a mobile, low-cost platform logging gas, dust and temperature along the rows is missing | A wireless sensor node on the robot reporting every 5 s to a logging dashboard |
| Disease classifiers are closed-set and report a disease for any input, even a wall | A vegetation gate that checks for leaf before the classifier is allowed to answer |
| Hobby quadruped designs give no guidance on centre of mass or the speed servos allow | Measured: centre of mass 12 mm aft, trot limited to about 0.18 m/s |

# Objectives and Scope

### Mobility

A low-cost printed 12-DOF quadruped whose gait is validated against real servo torque limits, then tested on the robot

### Sensing

Gas (TVOC, $eCO_2$), dust, temperature and range sensing carried on the chassis

### Plant-health AI

Leaf-disease classification that first checks a leaf is in view

### Live dashboard

Wireless logging, live charts, and leaf images tagged with the readings at capture

**In scope**

Printed platform, sensing and camera nodes, dashboard and classifier, locomotion design, simulation, first actuation trials

**Out of scope**

Trials on a working farm, sensor calibration against reference instruments, training a new classifier, GPS mapping

# 02

# Methodology and Design



How the system was designed: the workflow, key technical decisions, architecture, data flow and algorithms

# Methodology: Analyse, Revise, Build, Validate

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**01**

### Analyse

Torque, power and heat analysis of the Review III plan

**02**

### Revise

Move inference to a laptop; split sensing over two ESP32 nodes

**03**

### Build sensing

Sensor node and camera firmware, wiring with dividers

**04**

### Build software

Flask dashboard, logging, gated classifier

**05**

### Simulate

Kinematics and gait in MuJoCo with real servo limits

**06**

### Validate

Bench tests, logged sessions, scans, leg actuation

**Guiding principle** Every design change came from a measurement or analysis: supply and heat limits, servo torque margin, camera interface, and faults found on the bench.

# Key Technical Decisions Since Review III

| Review III plan | Final system | Reason |
|---|---|---|
| Jetson Orin Nano on the robot | XIAO ESP32-S3 and ESP32-CAM nodes; inference on a laptop | Needs 9–19 V and dissipates 7–25 W inside a PLA body; a fall risks an expensive board |
| IMX219 CSI camera | ESP32-CAM (OV3660), MJPEG over Wi-Fi | The ESP32-S3 has no MIPI-CSI camera interface |
| ROS 2 and MQTT on the robot | HTTP and JSON to a Flask dashboard | Fewer parts; every device on one local network |
| Gazebo dynamics model | MuJoCo physics model | The Gazebo stance never stabilised; MuJoCo walks and reports metrics |
| 3S LiPo and onboard regulators | USB power bank for the nodes; separate supply for servos | Safer and simpler for bench work |
| Walking trials on hardware | Gait in simulation first, then leg actuation on the robot | Simulation showed a centre-of-mass correction is needed |

# System Architecture

## Sensor node

XIAO ESP32-S3 + expansion board

- SGP30 gas · I²C 0x58
- MCP9808 temperature · I²C 0x18
- PPD42 dust · D0
- HC-SR04 × 2 · D6–D9
- OLED, buzzer, button

**JSON every 5 s**

## Laptop base station

Hotspot 192.168.137.1

- Flask dashboard on port 5000
- CSV log of every reading
- Vegetation gate + MobileNetV2
- Relays video to all viewers

**MJPEG video, 16–20 fps**

## Camera node

ESP32-CAM, OV3660 sensor

- MJPEG video on port 81
- Single photo on port 80
- Registers itself at boot

**Platform** SpotMicro-based printed chassis carrying both nodes · 12 × MG996R servos · gait validated in MuJoCo · any phone on the hotspot can open the dashboard

# Data Flow: From Sensor to Tagged Diagnosis

## Sensor stream

Sample: gas 1 s, temperature 2 s, dust 30 s

Post JSON to /api/sensors every 5 s

Grade good / moderate / high

Log to daily CSV, update live cards

## Camera stream

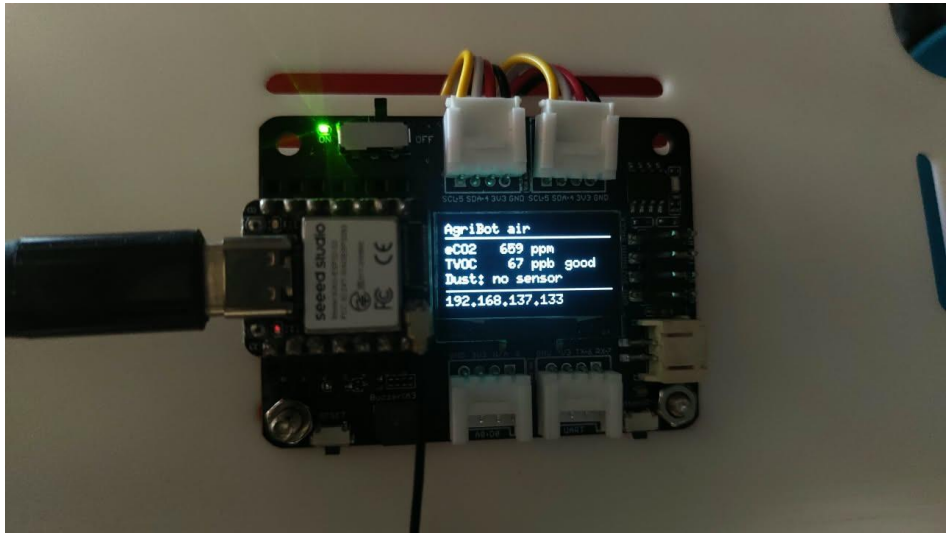
ESP32-CAM streams MJPEG

Dashboard relays frames to every viewer

Scan: vegetation gate, then MobileNetV2

Save image tagged with crop, condition and readings

# Sensor Node Hardware




*Live readings and IP address on the node's OLED*

**5 V → 3.3 V safely**
Every 5 V sensor output reaches the XIAO through a 1 kΩ / 2 kΩ divider: 5 V becomes 3.33 V, and the PPD42's 4 V becomes 2.67 V.

| Sensor | Measures | Interface | Status |
|---|---|---|---|
| SGP30 | TVOC, $eCO_2$ | $I^2C$ 0x58 | **Live** |
| MCP9808 | Temperature, ±0.25 °C | $I^2C$ 0x18 | **Live** |
| PPD42 | Particles > 1 µm | D0, via divider | **Live** |
| OLED, buzzer, button | Display and alerts | On board | **Live** |
| HC-SR04 #1 | Obstacle, row end | D6 / D7 | **Wiring pending** |
| HC-SR04 #2 | Canopy height | D8 / D9 | **Wiring pending** |

*Ultrasonic firmware is written and flashed; the sensors report as absent until connected.*

# Firmware Algorithms

### Dust by low-pulse occupancy

An interrupt times the PPD42's low pulses over 30 s. $C = 1.1r^3 - 3.8r^2 + 520r + 0.62$ particles per 0.01 $ft^3$

### Absent-sensor detection

An unplugged PPD42 read as 1.1 million particles. No edges and a low line now means absent

### Temperature-corrected range

$c = 331.3 + 0.606 \cdot T$ m/s from the MCP9808; a fixed 343 m/s is ~3 % off at 35 °C

### Crosstalk-free ultrasonics

The two HC-SR04s fire alternately, and each reading is a median of three

### Wi-Fi diagnostics

Reports why a link fails; traced −78 dBm to a missing antenna, −26 dBm once fitted

### Self-registering camera

The ESP32-CAM sends its stream URL to the dashboard at boot, so a new IP needs no setup

# Gated Leaf-Disease Classification

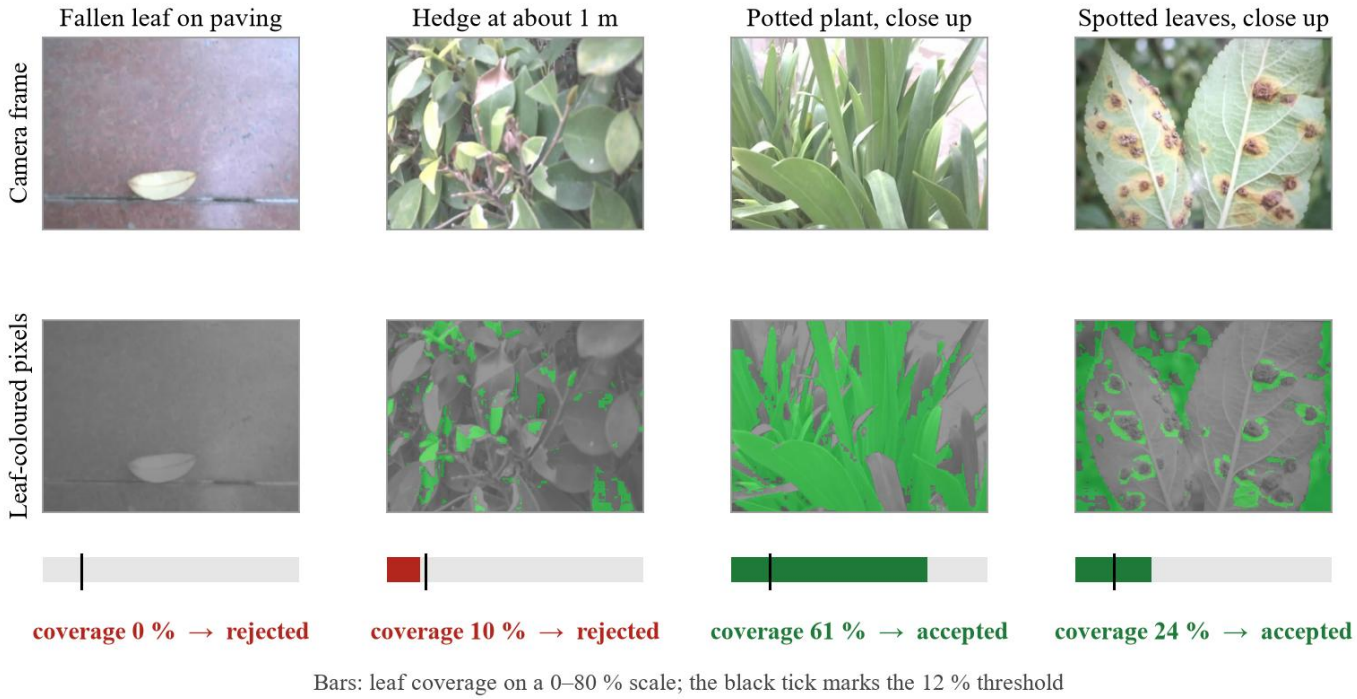
Frame from the ESP32-CAM

**Vegetation gate: share of pixels with 38° ≤ H ≤ 165°, S ≥ 0.20, V ≥ 0.12; below 12 % → no leaf**

MobileNetV2 trained on PlantVillage: 38 classes, 14 crops

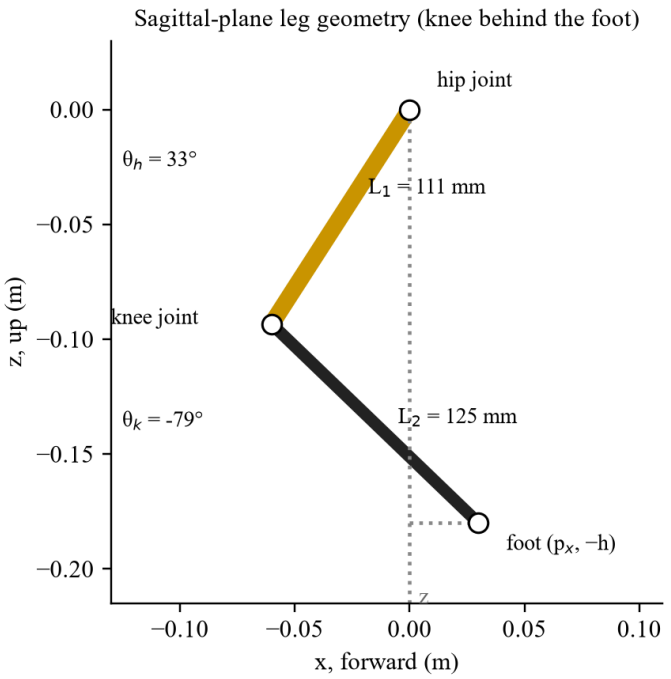
Top probability below 50 % → reported as uncertain

Image saved and tagged with crop, condition and the readings

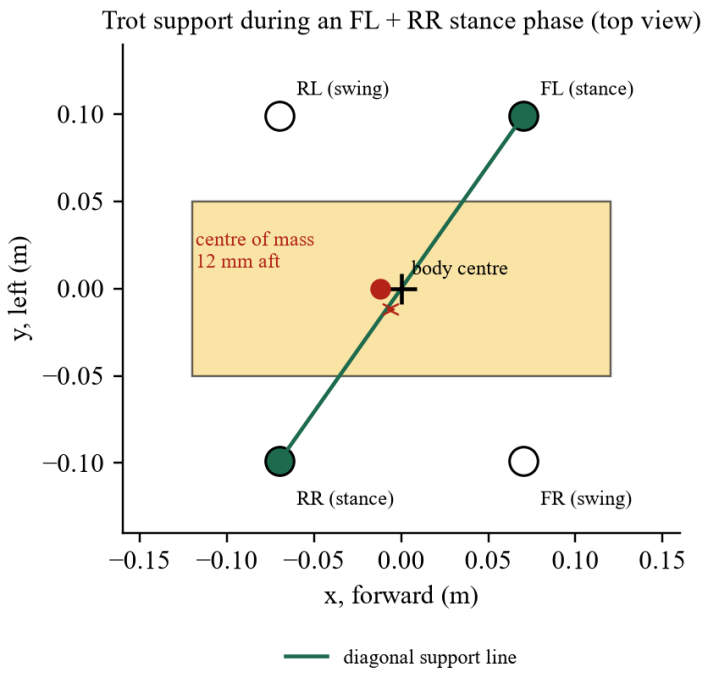



*The gate on four outdoor camera frames: leaf-coloured pixels in green*

# Locomotion Design: Kinematics, Gait, Centre of Mass




*Leg geometry: femur 111 mm, tibia 125 mm*




*Trot support line passes in front of the centre of mass*

### Closed-form IK

Abduction, hip, knee from the foot target; verified at 1,280 targets, error < $10^{-7}$ m

### Diagonal trot

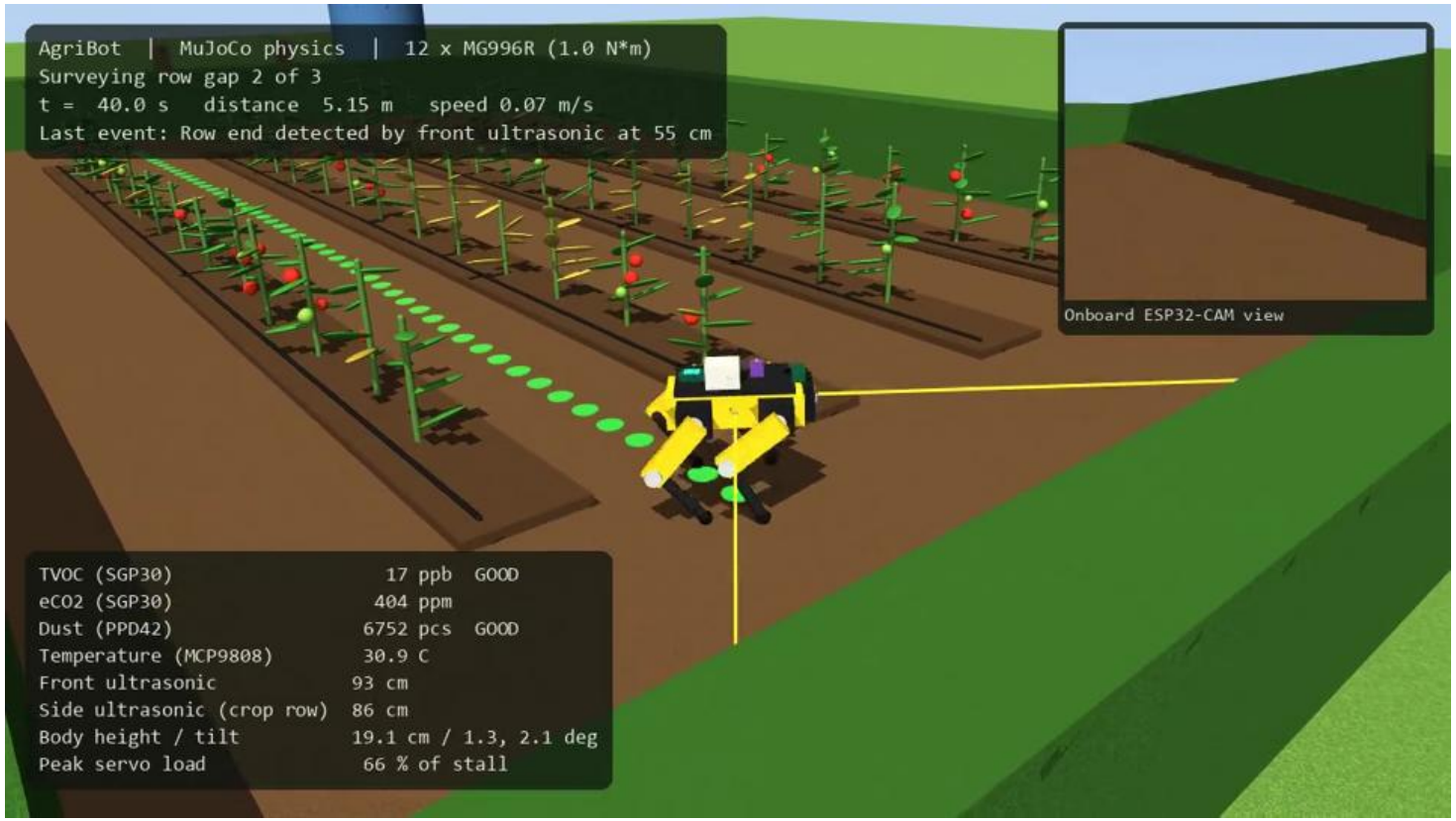
2 Hz, duty 0.6, 30 mm step height, 180 mm body height, roll and pitch levelling

### CoM-centred footholds

Feet placed around the measured centre of mass, 12 mm aft of the body centre

**Why footholds matter** Centred on the body, the trot fell in all 24 gait settings; centred on the measured centre of mass, tilt stayed below 0.7°.

# Simulation Model in MuJoCo




*Autonomous survey: stand up, trot each row gap, detect the row end, turn, log readings*

### Measured geometry

140 × 78 mm hip spacing, 111 / 125 mm legs, 1.35 kg

### Realistic servos

Position servos capped at 1.0 N·m (MG996R stall 0.92–1.08 N·m)

### Modelled field

4 tomato rows, a stressed patch emitting VOCs, a dusty headland

### Autonomous mission

Heading control; front range sensor ends each row at 55 cm

**Gait chosen by a sweep** 24 settings: frequency 2 / 2.5 / 3 Hz × duty 0.5 / 0.6 × stiffness 15 / 30 N·m/rad × levelling gain 0 / 0.5

# Technology Stack

## Hardware

XIAO ESP32-S3 + expansion board · ESP32-CAM (OV3660) · SGP30, MCP9808, PPD42, HC-SR04 × 2 · 12 × MG996R

## Firmware

Arduino-ESP32 core 3.3 · Adafruit SGP30 · U8g2 display · esp_http_server for MJPEG

## Dashboard and AI

Python · Flask · PyTorch · Hugging Face Transformers · MobileNetV2 (PlantVillage)

## Simulation

MuJoCo 3 · NumPy · Matplotlib · headless video and metrics

## Mechanical

SpotMicro base (CC-BY) · PLA+ and TPU · Python generators for the custom mounts

## Collaboration

Git and GitHub: firmware, dashboard, simulation and print files in one repository

# 03

# Completion and Integration



What was delivered, how the parts work together, and evidence of stable operation, followed by the live demo

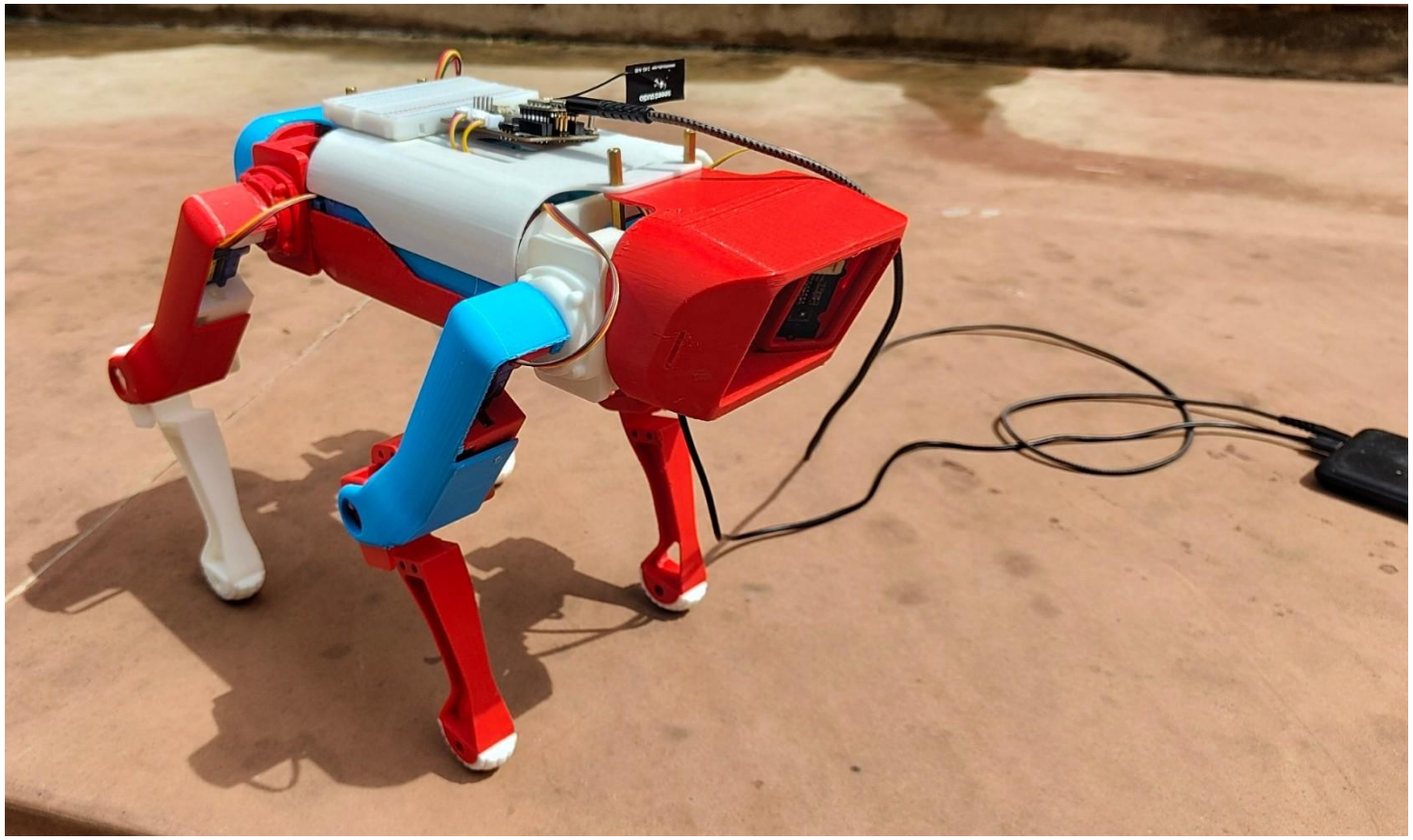
# Scope Delivered

**5 / 6**

objectives fully achieved

| Objective | Status | Evidence |
|---|---|---|
| Print and adapt the chassis; measure its geometry | **Achieved** | 35 parts printed; four custom parts; measured geometry |
| Wireless sensing payload, robust to faults | **Achieved** | Gas, dust, temperature live |
| Camera streaming and gated classification | **Achieved** | 16–20 fps; gate rejects non-leaf frames |
| Live dashboard with logging and tagging | **Achieved** | Logged sessions; tagged captures with readings |
| Verified kinematics; gait feasible in simulation | **Achieved** | IK error < $10^{-7}$ m; full survey, no fall |
| Walking on the physical robot | **Partly** | Legs actuate and it stands; no steps yet |

# The Assembled Robot



*The assembled robot standing outdoors*

### 35 printed parts

SpotMicro base adapted for the payload; ~700 g PLA+ with TPU feet; ₹5,200 to print

### Sensor node on the top cover

XIAO, breadboard and sensors, with the external antenna on a standoff

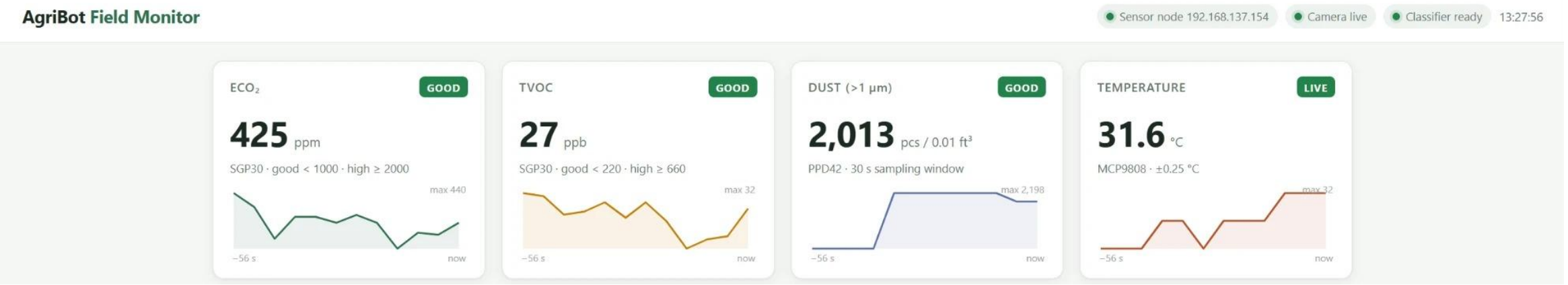
### Camera in the head

ESP32-CAM looks out through the printed nose

### No battery on board

Payload runs from a USB power bank beside the robot

# Integrated Dashboard




*Dashboard on 18 October 2026: live $eCO_2$, TVOC, dust and temperature from the sensor node*

### Status bands

TVOC 220 / 660 ppb (German UBA guideline); $eCO_2$ 1,000 / 2,000 ppm; node marked offline after 15 s of silence

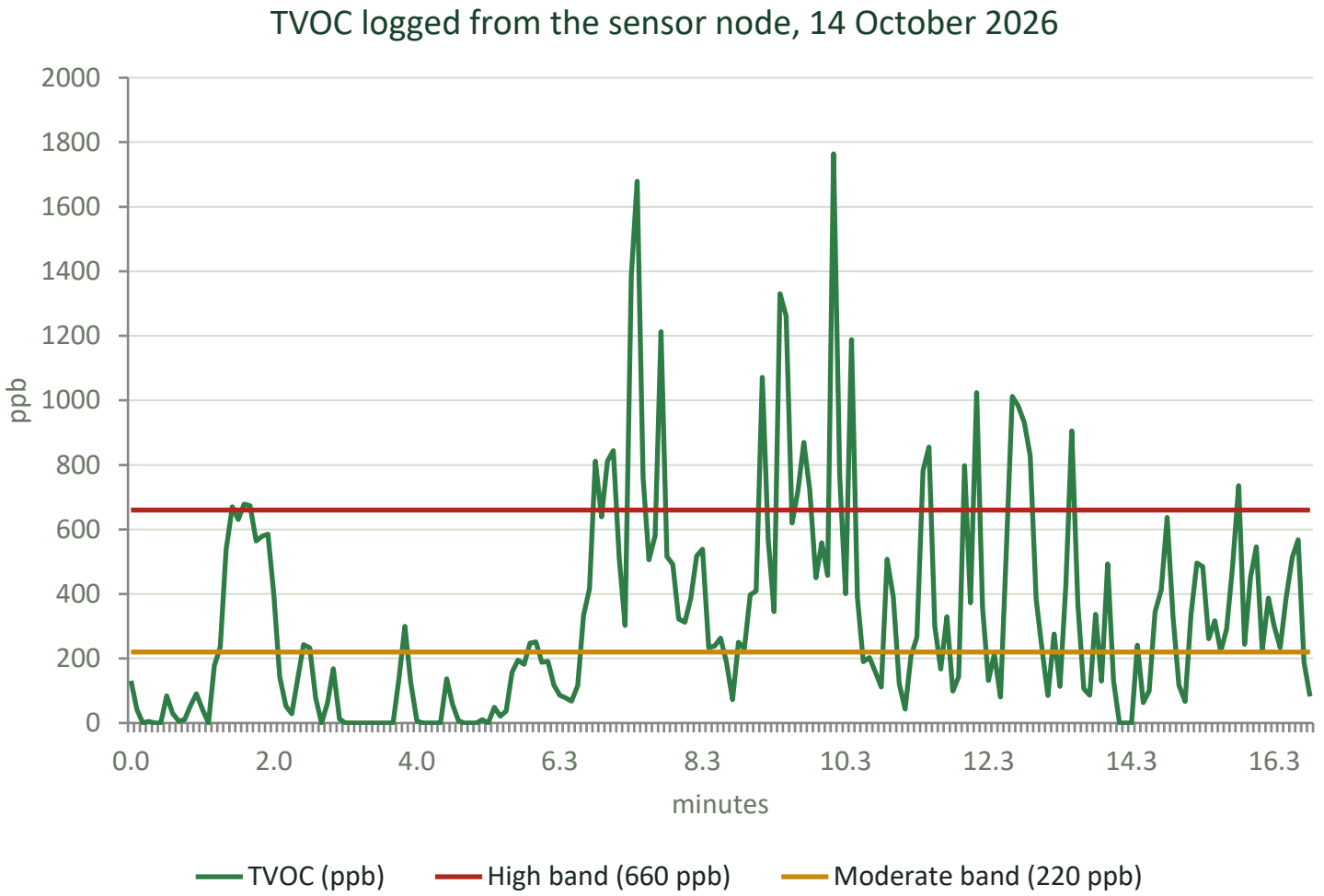
### Logging

Every reading to a daily CSV; every scan saved as a tagged image with the readings at capture

### Camera and diagnosis

Live MJPEG video, one-click scan or image upload, top three classes with confidence

# Stable Operation over Wi-Fi




**321 + 199**

readings in two sessions (28 and 17 min), no records lost

**1,764 ppb**

TVOC peak during exposure, back to baseline within tens of seconds

**−26 to −60**

dBm Wi-Fi signal throughout, ample for 5 s telemetry

# Live Demonstration

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**1** **Dashboard live**

Open http://192.168.137.1:5000; status pills show the sensor node, camera and classifier

**2** **Breath test**

Breathe on the SGP30: $eCO_2$ and TVOC rise within seconds and recover

**3** **Touch test**

Hold the MCP9808: temperature climbs about 3–4 °C

**4** **Leaf scan**

Show a leaf to the camera, then an empty scene: diagnosis versus no leaf in view

**5** **Simulation**

Run the MuJoCo survey: rows, turns, TVOC trail over the stressed patch

**6** **Robot**

Power the servos: the robot stands and moves its legs

# 04

# Testing and Results



Test procedures and measured results on the hardware, then the simulation metrics and their interpretation

# Bench Validation of the Hardware

| Test procedure | Measured result | What it shows |
|---|---|---|
| Breathe on the SGP30 | $eCO_2$ 400 → 746 ppm, TVOC 0 → 210 ppb within 5 s; back to baseline | Gas sensing responds and recovers |
| Finger on the MCP9808 | 31.5 → 35.0 °C in about 10 s | Temperature tracks a real change |
| PPD42 in room air | 0–5,065 particles / 0.01 $ft^3$ per 30 s window | Dust sensing live |
| PPD42 unplugged | Reported as absent instead of 1.1 million | No false alarms from wiring faults |
| Wi-Fi link of the XIAO | −78 dBm with time-outs → −26 dBm with antenna | Diagnostics found the fault |
| Camera video | ≈0.5 → 16–20 fps after moving to MJPEG | Live view usable for scanning |
| Servos, robot on its back | All four legs follow commanded angles | Wiring and servo control work |
| Servos, robot standing | Holds a stance and shifts its legs, slowly | Torque margin is thin, as predicted |

# Classifier Validation: The Gate Prevents False Diagnoses

**Without the gate**

## 82–90 %

confidence for “citrus greening” on a photo of an empty room

**With the gate**

## Rejected

0–0.1 % leaf coverage measured: “no leaf in view”, no diagnosis shown

## 12 / 12

PlantVillage leaves accepted by the gate (27–76 % coverage)

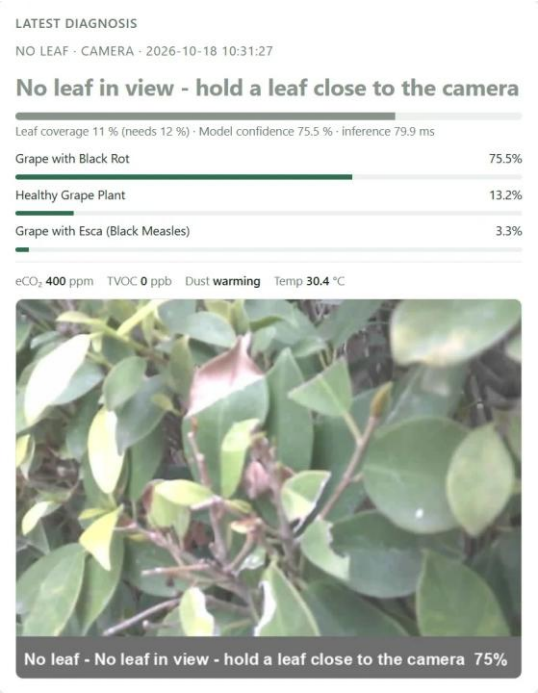
## 8 / 10

spot-check images classified correctly above 92 %; 1 uncertain, 1 wrong

## 30–80 ms

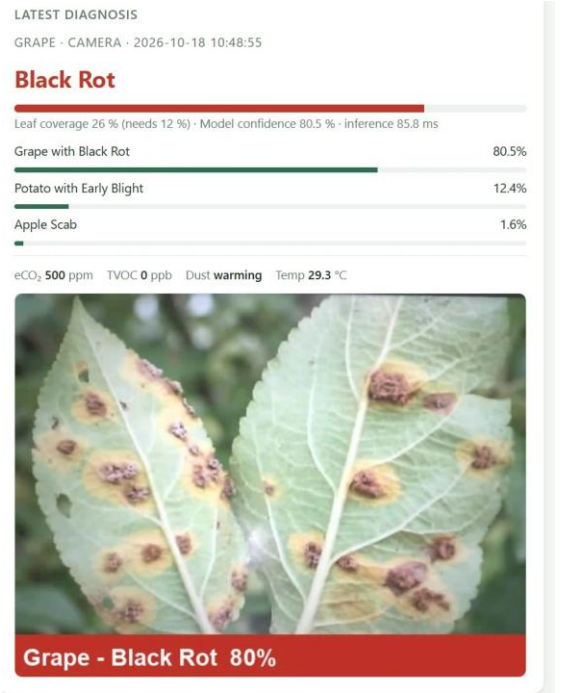
inference per image on the laptop CPU

# Live Scans Outdoors with the Robot's Camera










## Hedge at ~1 m: no leaf in view

11 % coverage, so the model's 75.5 % “black rot” is withheld

## Potted plant: low confidence

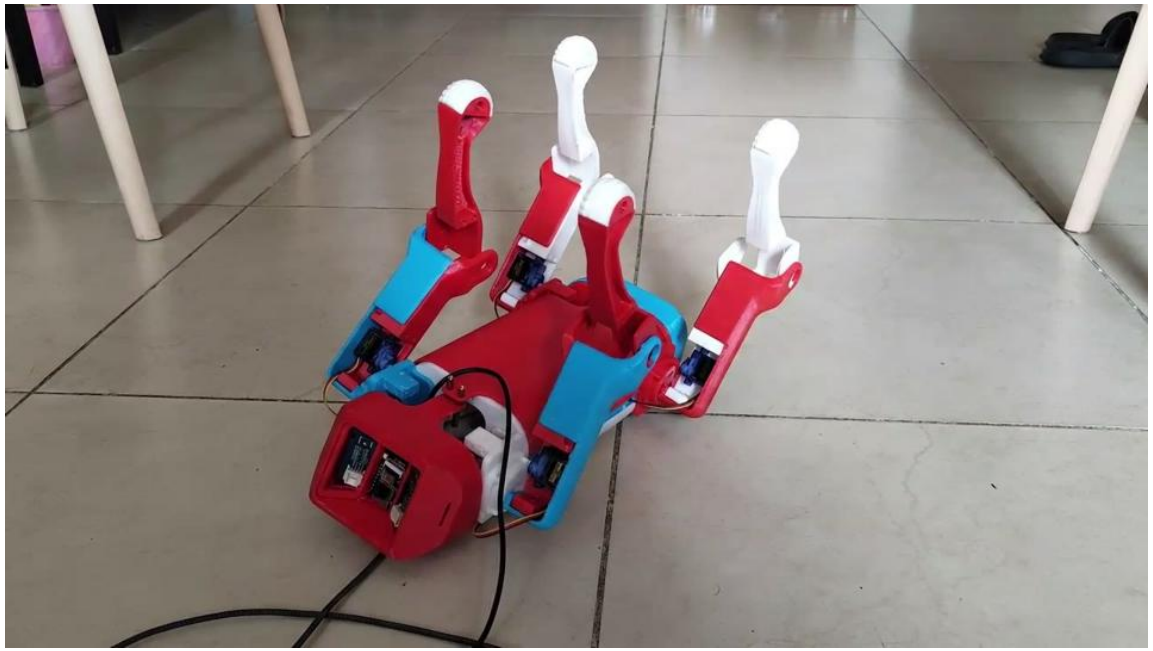
60 % coverage; “healthy corn” at 53.4 %, just above the 50 % threshold

## Spotted leaves: flagged

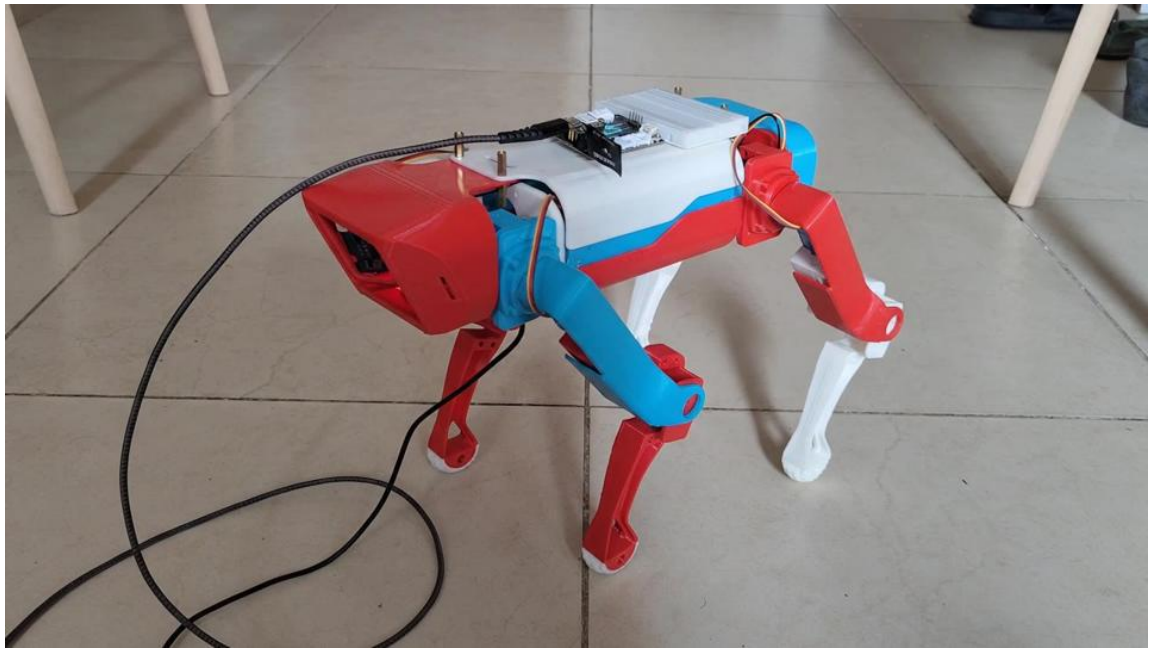
26 % coverage; “grape black rot” at 80.5 %, readings saved with the image

*Washed-out camera colours put some real foliage near the 12 % threshold; holding the leaf close fixes it.*

# Actuation Trials on the Physical Robot



*On its back: all four legs follow their commanded angles*



*Standing: supports its weight and shifts its legs, but weakly*

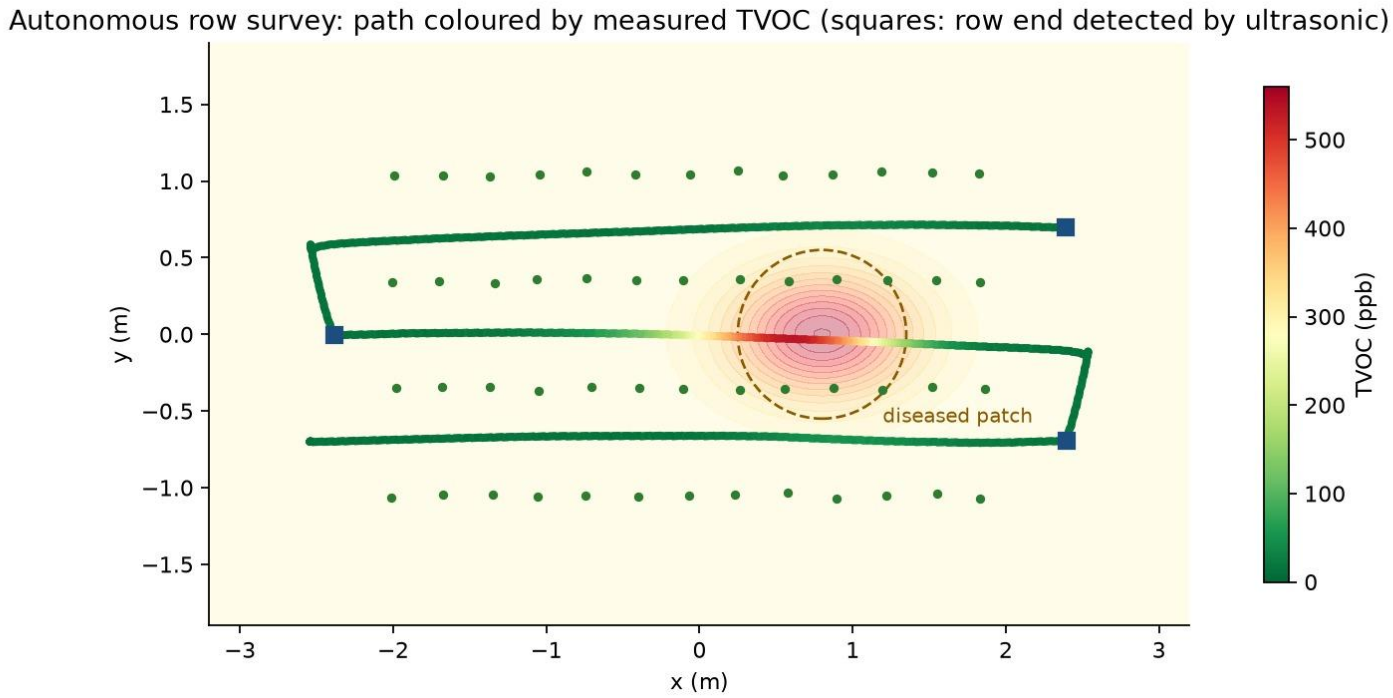
**Why it is weak**

Stance knee torque is 36–52 % of a genuine MG996R's rating and rises when a leg lifts; copies deliver less, and a sagging supply cuts torque further.

**Next step**

Dedicated 6 V servo supply, CoM-centred footholds and a lower stance; stronger servos if the margin is still thin.

# Simulated Autonomous Field Survey




*Route coloured by measured TVOC over the modelled VOC field; squares mark row ends detected by the front ultrasonic sensor*

**16.7 m**

in 129 s: all three row gaps, no falls

**3 / 3**

row ends detected by the ultrasonic sensor

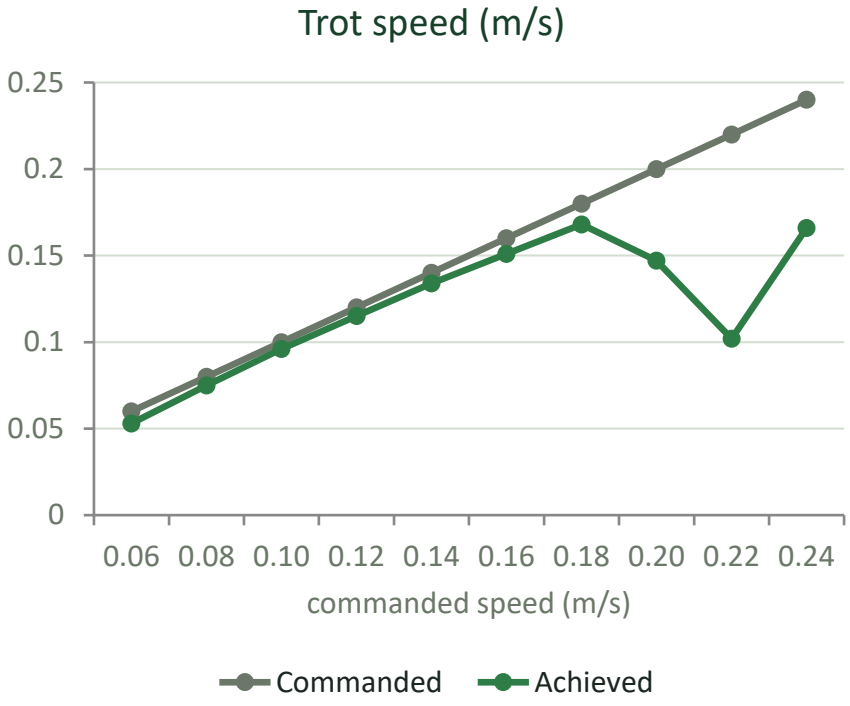
**≤ 2.7°**

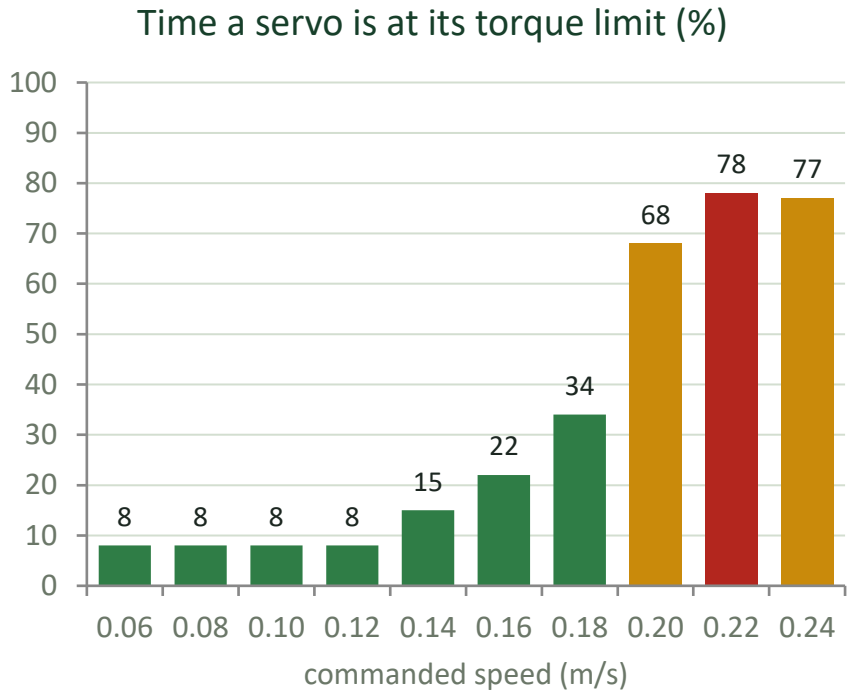
body tilt; a servo at its limit 1.3 % of the time

**535 ppb**

TVOC peak, 0.2 m from the source (sensor lag)

# Gait Stability and the Servo Speed Limit




**0.18 m/s**

usable ceiling with MG996R servos; tracks the command within 7 % from 0.08 m/s up

**Centre of mass 12 mm aft** With footholds centred on the body, all 24 gait settings fell within 0.8–3.0 s. Centred on the measured centre of mass, the trot held tilt below 0.7°.

# Outcome Against Review III Expectations

| Review III expectation | Outcome | Interpretation |
|---|---|---|
| Onboard inference on a Jetson | Inference on a laptop over Wi-Fi | Cheaper and safer; depends on the network link |
| Above 90 % disease accuracy (literature figure) | Functional demonstration; field accuracy not measured | Lab-trained models are expected to drop on field images |
| Static torque margin is thin | Confirmed in simulation (0.18 m/s ceiling) and on hardware | Analysis, simulation and robot agree |
| Walking robot with sensing | Robot stands and moves its legs; walking shown in simulation | Hardware walking is the next phase |
| Environmental sensing payload | Gas, dust, temperature live; ultrasonic firmware ready | Sensing payload essentially complete |

# 05

# Innovation and Relevance



What is new, whether it is feasible and affordable, how it is used responsibly, and where it goes next

# Contributions

### Legged platform with a crop-sensing payload

A printed 12-DOF quadruped carrying two Wi-Fi sensing nodes for gas, dust, temperature, range and video

### Vegetation gate for a closed-set classifier

A simple, explainable leaf check that stops confident disease reports for frames without a plant

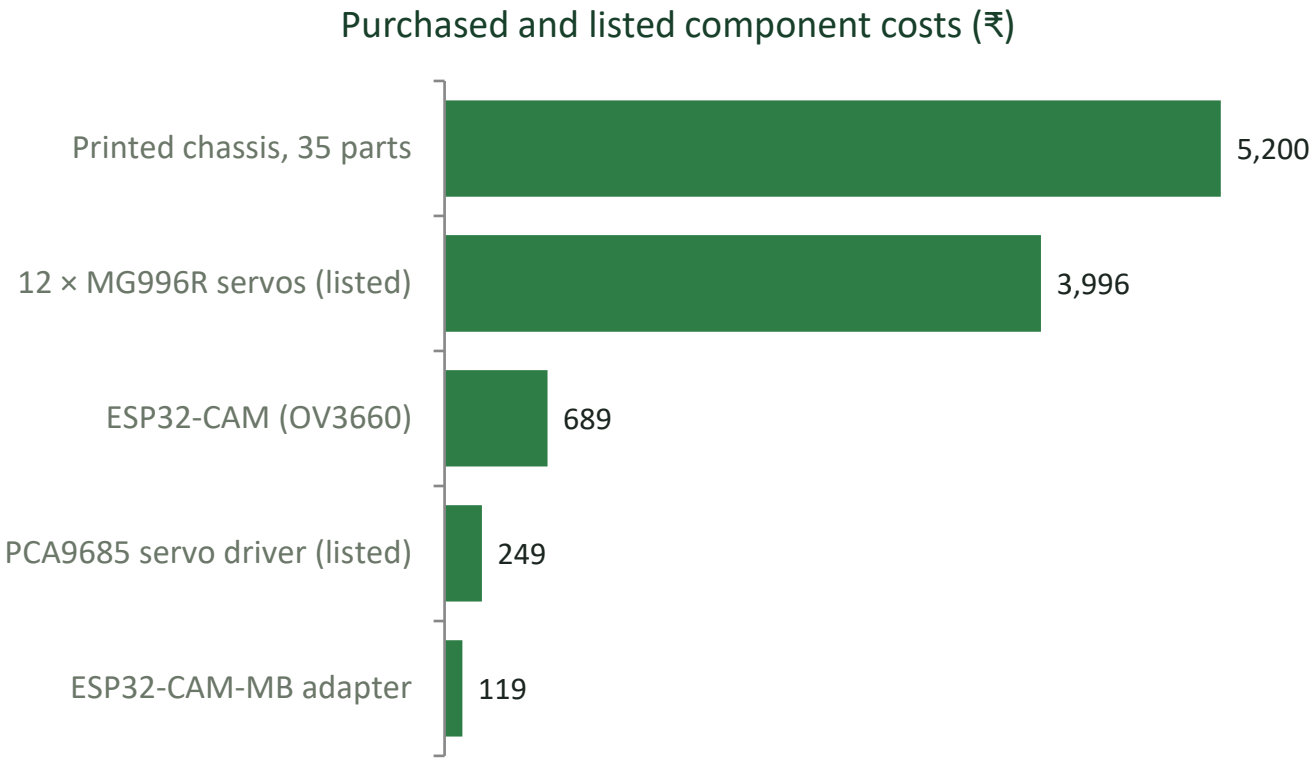
### Locomotion validated with real servo limits

A MuJoCo model from measured geometry, torque capped at MG996R stall, completing an autonomous row survey

### Design rules for printed quadrupeds

Centre footholds on the measured centre of mass (12 mm aft); expect about 0.18 m/s from MG996R servos

# Cost and Feasibility




*The XIAO, sensors, resistors and cables were already available to the team and are not costed.*

**≈ ₹10,250**

total purchased and listed cost of the platform

### Reported cost of research quadrupeds

| Platform | Reported cost |
|---|---|
| Stanford Doggo [2] | < US$3,000 |
| MIT Cheetah 3 [1] | ~US$50,000+ |
| ANYmal [4] | US$100,000+ |

# Ethical and Responsible Use

### Data stays local

All readings and images stay on the laptop's own hotspot; no cloud service, no personal data, camera aimed at crops

### No false certainty

Gate and 50 % threshold suppress unreliable labels; results are indications for an agronomist, not a spray decision

### Electrical and physical safety

Low-voltage USB payload, no LiPo on board, slow-moving robot, separate servo supply

### Honest sensor claims

$eCO_2$ is estimated and the dust sensor uncalibrated; used for trends and alerts, not regulatory values

### Open source and attribution

SpotMicro design credited under CC-BY; our code and print files are public

### Built for small farms

Low-cost parts and no internet needed, so the approach is within reach of smallholders

# Practical Applicability

### Row-by-row scouting

Walk the rows, log gas, dust and temperature, and scan suspicious leaves on the way

### Early-warning hotspots

Rising VOC readings along a row point to stressed plants before symptoms spread

### Nurseries and greenhouses

Repeated inspection of seedlings in narrow aisles where wheels struggle

### Records for agronomists

Tagged images with the conditions at capture give an evidence trail for advice

### What a farm needs to run it

- A laptop with Wi-Fi hotspot
- The robot with its two nodes
- A USB power bank
- Any phone to view the dashboard
- No internet, no cloud account

**Dashboard address**
http://192.168.137.1:5000

# Limitations

| Limitation | Effect | How it is handled |
|---|---|---|
| Real servos lack torque reserve | Robot stands and moves its legs but does not step | Dedicated supply, CoM footholds, lower stance, stronger servos |
| Classifier trained on lab photos, 14 crops | Lower accuracy on field images; any plant passes the gate | Gate and 50 % threshold; fine-tuning on field images |
| Washed-out ESP32-CAM colours | Some real foliage falls below the 12 % threshold | Hold the leaf close; tune the threshold to this camera |
| $eCO_2$ estimated; PPD42 uncalibrated | Relative readings, not regulatory values | Used for trends and alerts |
| Simulated fields are modelled | Simulation shows the method, not field data | Hardware sensors validated on the bench |

# Future Work

| | | |
|---|---|---|
| 01 | **Walking on hardware** | Dedicated 6 V servo supply; measure torque and voltage under load; CoM footholds and a lower stance; step in place, then walk below 0.18 m/s |
| 02 | **Complete the sensor set** | Wire both ultrasonic sensors; add a soil-moisture probe; calibrate gas and dust against reference instruments |
| 03 | **Act on measurements** | Grove relay to switch an irrigation pump or raise a gas alarm from the dashboard |
| 04 | **Field-ready classification** | Fine-tune on images from the robot's camera; replace the colour gate with a learned open-set detector |
| 05 | **Mapping and autonomy** | Compensate for the SGP30's lag; log positions to build field maps; battery for untethered runs |

# Conclusion

### Sensing works end to end

Two wireless nodes, a live dashboard with logging, and leaf-disease classification that refuses to guess without a leaf

### Locomotion validated in physics

A full autonomous row survey without a fall, with servos capped at their real torque

### The printed robot moves

All twelve servos drive the legs and it stands, with too little torque reserve to step, as predicted

### Design rules for the next build

Centre footholds on the centre of mass; expect about 0.18 m/s from MG996R servos

**Repository** github.com/AdithyaSM31/Precision-Agriculture-Monitoring-Quadruped-Robot-Dog

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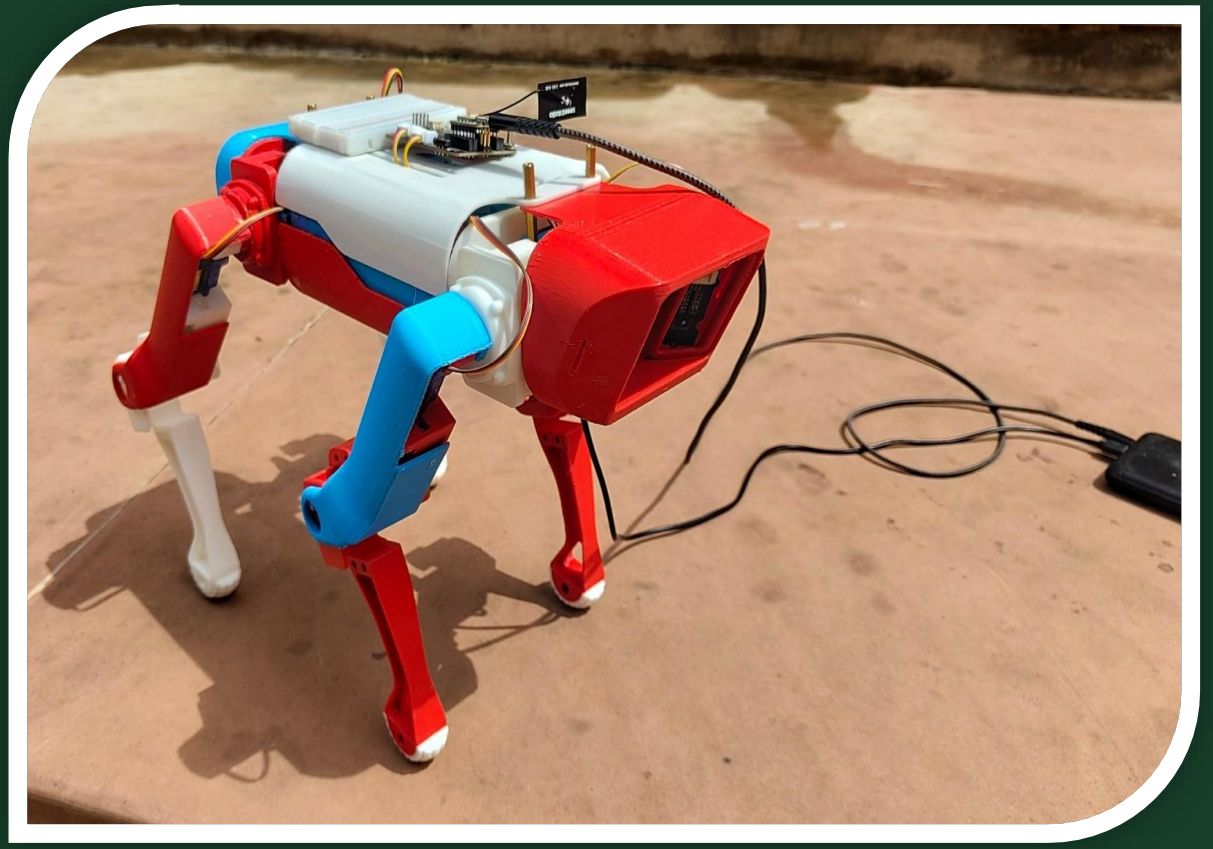
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# Thank you

Questions and live demonstration



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# Literature Review, Papers 1–6

| # | Authors and year | Title | Key contribution | Research gap |
|---|---|---|---|---|
| 1 | Bledt et al. (2018) | MIT Cheetah 3: Design and Control of a Robust, Dynamic Quadruped Robot | Designed high-torque density proprioceptive actuators and Model Predictive Control (MPC) for dynamic walking over rough terrain without vision. | High cost (~$50k+), complex custom planetary gearing, no environmental/agricultural sensing payload. |
| 2 | Kau et al. (2019) | Stanford Doggo: An Open-Source, Quasi-Direct-Drive Quadruped | Developed an open-source 3D-printed quadruped platform using quasi-direct drive brushless motors for dynamic bounding. | Optimized purely for acrobatic jumps/speed; lacks on-board compute for perception and has <10 min battery life. |
| 3 | Katz et al. (2019) | Mini Cheetah: A Hardware Design for Extremity Locomotion | Modular 3-DOF leg design using low-cost backdrivable BLDC motors enabling agile omnidirectional gaits. | Focuses strictly on mechanical agility in laboratory environments; no autonomous navigation or edge AI pipeline. |
| 4 | Hutter et al. (2016) | ANYmal - A Unique Quadruped Robot for Autonomous Operation in Harsh Environments | Industrial-grade quadruped with series elastic actuators (SEAs) and IP67 ruggedized body for outdoor inspection. | Prohibitively expensive ($100k+), heavy industrial form-factor unsuitable for smallholder precision farming. |
| 5 | Ding et al. (2021) | Learning to Walk on Uneven Terrain for Quadrupedal Robots using Reinforcement Learning | Trained neural network locomotion policies in simulation (Isaac Gym) with robust sim-to-real transfer over rugged slopes. | High computational training overhead; policy does not integrate multi-sensor agronomic payloads or path tracking. |
| 6 | Bellicoso et al. (2018) | Dynamic Locomotion Through Online Nonlinear Model Predictive Control for Quadrupedal Robots | Real-time ZMP and CoM trajectory optimization enabling adaptive crawl and trot gaits over unmodeled obstacles. | Requires high-end x86 computer boards; difficult to run concurrently with edge vision models on low-power platforms. |

# Literature Review, Papers 7–12

| # | Authors and year | Title | Key contribution | Research gap |
|---|---|---|---|---|
| 7 | Fehlmann et al. (2023) | Precision Agriculture Using Autonomous Ground Robots: A Review | Systematic review of wheeled and tracked agricultural rovers for weed control, phenotyping, and soil sampling. | Identifies severe mobility failure of wheeled rovers in wet mud, soft soil, and high-crop canopy furrows. |
| 8 | Roldán et al. (2018) | Mini-UAVs and Mobile Robots for Precision Agriculture: A Review of Multi-Robot Systems | Analyzed complementary roles of aerial drones (macro-surveys) and ground robots (micro-inspection). | Highlighted drone limitations: 20-min flight time, inability to inspect leaf undersides, and blind spots beneath crop canopies. |
| 9 | Bogue, R. (2016) | Robots in Agriculture: State of the Art and Future Directions | Surveyed commercial adoption of agricultural robotics, highlighting high ROI in crop health monitoring. | Commercial units are bulky, heavy, cause soil compaction, and require flat structured greenhouse tracks. |
| 10 | Vasconez et al. (2020) | Human–Robot Interaction in Agriculture: A Survey and Review | Investigated telemetry, UX dashboards, and safety mechanisms for farmer-robot interfaces in open fields. | Found most agricultural robotic interfaces are academic prototypes lacking real-time edge telemetry and alert maps. |
| 11 | Mohanty et al. (2016) | Using Deep Learning for Image-Based Plant Disease Detection | Trained AlexNet and GoogLeNet on 54,306 images across 26 diseases with 99.35% classification accuracy. | Tested only on static laboratory-controlled leaf images with uniform backgrounds; lacks edge device deployment. |
| 12 | Ferentinos, K. P. (2018) | Deep Learning Models for Plant Disease Detection and Diagnosis | Evaluated VGG, ResNet, and AlexNet on 87,848 leaf images with real-world noise variations (99.5% accuracy). | High parameter count (VGG/ResNet) causes severe latency and memory bottlenecks on embedded single-board computers. |

# Literature Review, Papers 13–18

| # | Authors and year | Title | Key contribution | Research gap |
|---|---|---|---|---|
| 13 | Too et al. (2019) | A Comparative Study of Fine-Tuning Deep Learning Architectures for Plant Disease Identification | Benchmarked MobileNetV2, DenseNet-121, and InceptionV3; proved MobileNetV2 achieves 97.4% accuracy with 10× fewer FLOPs. | Did not perform hardware-specific quantization (TensorRT / ONNX) or test real-time video stream ingestion. |
| 14 | Chen et al. (2020) | Using Deep Transfer Learning for Image-Based Plant Disease Identification on Embedded Devices | Deployed lightweight CNN on Raspberry Pi 3B+ using INT8 quantization, achieving 12 FPS at 92.1% accuracy. | Insufficient FPS for continuous mobile robot navigation; no integration with robotic gait or sensor fusion. |
| 15 | Saleem et al. (2019) | Plant Disease Detection and Classification by Deep Learning: A Review | Comprehensive meta-analysis of deep learning architectures, dataset biases, and real-time field challenges. | Concluded that field-ready solutions require mobile ground robots that bring cameras within 0.5m of plant leaves. |
| 16 | Ray, P. P. (2017) | Internet of Things for Smart Agriculture: Technologies, Practices and Future Direction | Proposed layered IoT architecture for soil moisture, microclimate, and particulate monitoring in precision farming. | Relied exclusively on stationary sensor nodes (motes), creating spatial coverage gaps across large farm tracts. |
| 17 | Elijah et al. (2018) | An Overview of Internet of Things (IoT) and Data Analytics in Smart Agriculture | Investigated wireless sensor networks (WSNs) and cloud data pipelines for environmental anomaly detection. | High infrastructure cost of deploying hundreds of stationary nodes; vulnerable to field vandalism and weather damage. |
| 18 | Bacco et al. (2019) | Smart Farming: Technologies, Applications and Challenges in Precision Agriculture | Explored multi-sensor data fusion (temperature, humidity, organic gases, optical imagery) for soil health profiling. | Lacked mobile autonomous platforms to transport multi-sensor payloads dynamically throughout crop rows. |